

Training-Budget Trade-offs in a Standard Recommender

Decision memo for a conventional BPR-SGD benchmark on MovieLens latest-small.

Decision

Use **5 epochs / 1 negative** for routine iteration. Use **5 epochs / 3 negatives** for a quality-focused final run when +0.016 NDCG@10 and +2.08 percentage points HR@10 justify 2.88x the measured training time.

Evidence

Setting	Updates	HR@10	NDCG@10	Time	Top-10 agreement
2 ep / 1 neg	95,942	0.630	0.385	1.89 s	0.860
5 ep / 1 neg	239,855	0.636	0.388	4.57 s	0.898
5 ep / 3 neg	719,565	0.657	0.404	13.18 s	0.870
10 ep / 1 neg	479,710	0.641	0.391	9.21 s	0.897

Metrics are means across three optimizer seeds. Top-10 agreement is the mean Jaccard overlap across seed pairs on fixed evaluation candidate sets. Runtime covers model training only and is machine-dependent.

Interpretation

The routine setting gives the strongest mean top-10 agreement (0.898) while keeping training time to 4.57 seconds. Sampling three negatives per positive produces the best held-out ranking metrics, but its 13.18-second runtime is 2.88 times the routine setting. More epochs with one negative cost more without matching that quality gain in this experiment.

Limits

This is one chronological holdout on a small development dataset with 99 sampled evaluation negatives per user. It does not compare production systems, tune hyperparameters, or establish a universal stopping rule. Repeat runtime measurements and define a business-specific quality threshold before applying the recommendation elsewhere.

Method: standard implicit-feedback matrix factorization trained with BPR logistic SGD. Public data: [MovieLens latest-small](#) (100,836 ratings, 610 users, 9,742 movies). Raw data are downloaded at runtime and are not redistributed.